YOKESH D

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Profile Summary

Final-year M.Tech Integrated Software Engineering student specializing in AI-driven application development, with hands-on experience building RAG pipelines, multi-agent AI systems, NLP-based classification systems, and cloud-deployed ML solutions. Proficient in Python, LangChain, BERT, PyTorch, FastAPI, and AWS, with a strong track record of taking projects from model development to production deployment using Docker and CI/CD pipelines.

Technical Skills

- **Programming Languages:** Python, Java
- **Databases:** MySQL, Firebase
- **Data Engineering:** Databricks, Apache Spark, Delta Lake
- **Cloud & DevOps:** AWS (EC2, S3, CloudWatch, SNS, IAM), Docker, CI/CD, Jenkins
- **AI & Machine Learning:** BERT, NLP, RAG, LangChain, PyTorch, OpenCV, EfficientNetB3, CNN, SHAP, FAISS
- **Tools:** Git, GitHub, Postman, n8n, VS Code, Google Colab
- **Core Concepts:** DSA, OOP, DBMS, Operating Systems, SDLC, Computer Networks, REST APIs

Projects

HealthSense AI – Early Detection of Neurodegenerative Diseases using Explainable AI & RAG **GitHub**

Technologies: Python, FastAPI, React, TypeScript, scikit-learn, SHAP, FAISS, SentenceTransformers, Groq LLM, PubMed API

- Built a supervisor-driven multi-agent clinical decision-support platform for Alzheimer's and Parkinson's risk analysis, combining disease-specific prediction, questionnaire screening, SHAP explainability, biomedical RAG, and PubMed research retrieval.
- Trained Random Forest models on ADNI and PPMI clinical data, achieving 91.57% accuracy (ADNI, 3-class) and 92.42% accuracy with 97.81% ROC-AUC (PPMI); used SHAP TreeExplainer for feature-level interpretation.
- Implemented a RAG pipeline with all-MiniLM-L6-v2 embeddings and FAISS over 2,514 indexed biomedical chunks, with LLM synthesis (Groq) and response validation; verified with 12/12 passing backend tests. Authored a research paper on the system.

SafeText AI – Intelligent Toxic Content Detection System **GitHub**

Technologies: Python, BERT, NLP, PyTorch, FastAPI, Docker, GitHub, Jenkins, CI/CD, Oracle Cloud (OCI), Gradio

- Developed an AI-powered toxic comment moderation system using BERT to classify harmful content including toxicity, threats, insults, and identity-based hate speech.
- Built a production-ready ML pipeline with FastAPI backend, preprocessing workflow, Docker containerization, and Jenkins-based CI/CD automation.
- Deployed the containerized application on Oracle Cloud Infrastructure (OCI), enabling scalable real-time toxicity detection through an interactive Gradio interface.

Cloud Honeypot with Real-Time Attack Detection & Alert System

Technologies: AWS EC2, CloudWatch, SNS, S3, Linux

- Designed and deployed a cloud-based honeypot system on AWS for real-time cyber-attack monitoring and logging.
- Integrated CloudWatch, SNS, and S3 for automated attack alerts, monitoring, forensic log storage, and security analytics.
- Implemented Fail2Ban and custom monitoring scripts for closed-loop automated threat response, capturing and analysing 138+ attack events — demonstrating real-time network behaviour monitoring and corrective action recommendation.

Education

Vellore Institute of Technology, Vellore 2022 – Present

M.Tech (Integrated) Software Engineering | CGPA: 7.84 / 10

AIM Matric Higher Secondary School, Arni 2020 – 2022

Higher Secondary (HSC) | Grade: 92.1%

Training & Certifications

- AWS Certified Cloud Practitioner (CLF-C02) — Amazon Web Services, 2026
- Fundamentals of AI Agents Using RAG and LangChain — IBM, 2026
- Generative AI Bootcamp — VIT, 2025